

Cascade Calculation Methods in hydrocal/src/CoolerFractions.cxx

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1 Problem

An ion beam is created in hydrogenic excited states $|n, l\rangle$ and traverses a cooler section of length L before entering a field ioniser. During the flight of duration $t = L/v_{\text{ion}}$ the states undergo radiative cascade. We need the *detection probability*

$$f_{nl} = \sum_{n', l'} [\exp(\mathbf{M} t)]_{n' l', nl} p_{\text{surv}}(n', l') \quad (1)$$

where $p_{\text{surv}}(n, l)$ is the field-ionisation survival probability (0 or 1 below/above the field-ionisation threshold n_{zero}) and $\mathbf{M}$ is the radiative rate matrix.

2 Rate Matrix

States are indexed as $(n, l) \rightarrow i = n(n-1)/2 + l$, giving $N = n_{\text{max}}(n_{\text{max}}+1)/2$ states. The rate matrix $\mathbf{M}$ is upper-triangular (decays only go to lower n):

$$M_{ii} = -\frac{1}{\tau_i}, \quad M_{ji} = \frac{B(i \rightarrow j)}{\tau_i} \quad (j < i) \quad (2)$$

where τ_i is the radiative lifetime of state i and $B(i \rightarrow j)$ the branching ratio for the dipole transition $i \rightarrow j$. Column sums of $\mathbf{M}$ vanish, guaranteeing probability conservation.

3 Method 1: Matrix Exponential

The population vector obeys $d\mathbf{p}/dt = \mathbf{M} \mathbf{p}$. Its formal solution $\mathbf{p}(t) = \exp(\mathbf{M} t) \mathbf{p}(0)$ gives the exact cascade evolution for all coupled states simultaneously.

3.1 Implementation

The matrix exponential is computed by scaling and squaring with a Padé [13/13] approximant [1]:

1. Compute the 1-norm $\|\mathbf{M} t\|_1$ and determine a scaling parameter $s = \max(0, \lceil \log_2(\|\mathbf{M} t\|_1/2) \rceil)$.
2. Form $\mathbf{Q} = \mathbf{M} t / 2^s$.
3. Evaluate the Padé numerator $\mathbf{N}(\mathbf{Q})$ and denominator $\mathbf{D}(\mathbf{Q})$ polynomials of degree 13 using Horner's rule on even/odd powers of $\mathbf{Q}^2$.
4. Solve $\mathbf{D} \mathbf{R} = \mathbf{N}$ by Newton iteration $\mathbf{R}_{k+1} = \mathbf{R}_k(2\mathbf{I} - \mathbf{D} \mathbf{R}_k)$, starting from $\mathbf{R}_0 = \mathbf{I}$. Three iterations suffice ($\|\mathbf{I} - \mathbf{D}\|$ is small after scaling). Then $\mathbf{R} \leftarrow \mathbf{R} \mathbf{N}$ to obtain $\mathbf{R} = \mathbf{D}^{-1} \mathbf{N} \approx \exp(\mathbf{Q})$.

5. Square s times: $\mathbf{R} \leftarrow \mathbf{R}^2$ to recover $\exp(\mathbf{M} t)$.

Finally, f_{nl} is obtained from Eq. (1) as a single matrix–vector product $\mathbf{R}^\top p_{\text{surv}}$. For an extended creation region ($x_1 \neq x_2$) a trapezoidal average of $\exp(\mathbf{M} t_1)$ and $\exp(\mathbf{M} t_2)$ is used.

3.2 Complexity

Matrix multiplication dominates at $O(N^3)$ per multiply. For $n_{\text{max}} = 20$ ($N = 210$): the scaling factor is $s \approx 15$, giving roughly 20 matrix multiplications in total. The $O(N^3)$ cost is modest for $N \lesssim 1000$ but becomes prohibitive for very large n_{max} .

4 Method 2: Recursive Cascade

The original `hydrocal` recursive method builds f_{nl} analytically, state by state, without ever forming the full matrix [2].

4.1 Core formula

For an initial state (n_1, l_1) the detection probability is

$$f_{n_1 l_1} = (1 - p_d(n_1 l_1)) p_{\text{surv}}(n_1 l_1) + \sum_{n_2, l_2} B(n_1 l_1 \rightarrow n_2 l_2) \left[(p_d(n_1 l_1) - p_{\text{casc}}) p_{\text{surv}}(n_2 l_2) + f_{\text{casc}} \right] \quad (3)$$

where $p_d(nl) = 1 - e^{-t/\tau_{nl}}$ is the single-state decay probability and p_{casc} is the effective cascade-chain decay probability computed by the Lagrange-type identity:

$$p_{\text{casc}} = \sum_{i=0}^K \frac{p_{d,i}}{\prod_{k \neq i} (1 - \tau_k / \tau_i)} \quad (4)$$

for a chain of $K+1$ states with lifetimes $\tau_0, \tau_1, \dots, \tau_K$ and individual decay probabilities $p_{d,i}$. The term f_{casc} is the recursive contribution from deeper cascade steps, computed by calling `cascade()` with the chain extended by one state.

The recursion depth is controlled by `ncascstep`:

- `ncascstep = 0`: no cascade (direct survival only).
- `ncascstep > 0`: at most that many cascade steps.
- `ncascstep < -1`: unlimited ($= n_{\text{max}}$).
- `ncascstep = -1`: selects the matrix exponential method.

4.2 Complexity

The recursion visits each (n, l) state a limited number of times. The total work scales roughly as $O(N \times \langle n_{\text{cascade}} \rangle \times n_{\text{neighbours}})$, where $\langle n_{\text{cascade}} \rangle$ is the average chain length and $n_{\text{neighbours}} \leq 2$ (dipole selection rules allow at most two l -values per n_2). For $n_{\text{max}} = 20$ this is very fast ($\sim 10^5$ operations).

5 Comparison

Both methods were verified against each other for $n_{\text{max}} = 20$, $q_{\text{ion}} = 3$, and realistic CSRe parameters ($v_{\text{ion}} = 2.0 \times 10^8$ cm/s, $L = 68$ cm). The maximum absolute difference across all 210 states is $\sim 3 \times 10^{-7}$, consistent with accumulated floating-point round-off in two independent algorithms.

	Matrix exponential	Recursive cascade
Mathematical basis	Exact ODE solution	Lagrange interpolation identity
Complexity	$O(N^3)$	$O(N \times n_{\text{chain}})$
Matrix storage	$N \times N$	$O(N)$
Arbitrary n_{max}	Expensive for large N	Always fast
Exactness	Machine precision	Machine precision
Truncation control	None (exact)	Via <code>ncascstep</code>
Extended creation	Trapezoidal average	Trapezoidal average

Table 1: Comparison of the two cascade calculation methods.

5.1 Recommendations

- For $n_{\text{max}} \leq 200$ both methods are fast. The matrix exponential is simpler to implement and verify, and gives the exact answer in one shot.
- The recursive method is preferred when n_{max} is large ($\gtrsim 500$), since its cost scales linearly in N rather than cubically. The `ncascstep` parameter provides a natural convergence knob.
- Both methods handle the trapezoidal position average for extended creation regions identically.

References

- [1] N. J. Higham, *Functions of Matrices: Theory and Computation*, SIAM - Society for Industrial and Applied Mathematics, Philadelphia, USA (2008).
- [2] S. Schippers, A. Müller, G. Gwinner, J. Linkemann, A.A. Saghir, and A. Wolf, *Storage ring measurement of the C IV recombination rate coefficient*, *Astrophys. J.* **555**, 1027 (2001).